Notes

November 12, 2020

Selected Solutions

§1.2 The framework of Riemannian geometry

1. Consider the line element $ds^2 = \dfrac{dx^2 + dy^2}{y^2}$ on the upper half-plane $y > 0$.

Proof. We compute the length of a parametrized path joining $p_0 = (-1, 1)$ at $t = a$ and $p_1 = (1, 1)$ at $t = b$ using the line element as follows:

$$\begin{aligned} L(\gamma) &= \int_{p_0}^{p_1} ds = \int_{p_0}^{p_1} +\sqrt{ds^2} = \int_{p_0}^{p_1} \sqrt{\frac{dx^2 + dy^2}{y^2}} = \int_{p_0}^{p_1} \sqrt{\frac{dx^2}{y^2} + \frac{dy^2}{y^2}} \\ &= \int_a^b \sqrt{\frac{1}{y^2}\left[\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2\right] dt^2} = \int_a^b \frac{dt}{y(t)} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}. \end{aligned}$$

(a) Let c_1 be the "straight" line segment:

$$\gamma_1(t) = (x(t), y(t)), \qquad x(t) = t, \qquad y(t) = 1, \qquad a = -1 \le t \le 1 = b,$$

whose length is computed below:

$$L(\gamma_1) = \int_{-1}^{1} \frac{dt}{1} \sqrt{(1)^2 + (0)^2} = \int_{-1}^{1} dt = [t]_{-1}^{t=1} = 1 - (-1) = 2.$$

(b) Let c_2 be the "circular arc":

$$\gamma_2(t) = (x(t), y(t)), \qquad x(t) = \sqrt{2}\cos(t), \qquad y(t) = \sqrt{2}\sin(t), \qquad \frac{\pi}{4} \le t \le \frac{3\pi}{4},$$

whose length is similarly computed below:

$$\begin{aligned} L(\gamma_1) &= \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{dt}{\sqrt{2}\sin(t)} \sqrt{\left(-\sin(t)\sqrt{2}\right)^2 + \left(\cos(t)\sqrt{2}\right)^2} = \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{\sqrt{2}\, dt}{\sqrt{2}\sin(t)} \sqrt{1} \\ &= \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{dt}{\sin(t)} = \left[-\ln\left(\cot(t) + \csc(t)\right)\right]_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \\ &= -\left[\ln\left(\cot\left(\frac{3\pi}{4}\right) + \csc\left(\frac{3\pi}{4}\right)\right) - \ln\left(\cot\left(\frac{\pi}{4}\right) + \csc\left(\frac{\pi}{4}\right)\right)\right] \\ &= -\left[\ln\left(-1 + \sqrt{2}\right) - \ln\left(+1 + \sqrt{2}\right)\right] = \sinh^{-1}(1) - (-\sinh^{-1}(1)) \\ &\approx 1.762747174039086\ldots. \end{aligned}$$

Thus, the circular arc is shorter than the straight line segment. This appears to be the Poincaré upper halfspace model of the 2-dimensional hyperbolic[1] space $\mathbb{H}^n 2$, as discussed in the next section. □

[1] (Satire) The reader is encouraged to view this video on Lobachevsky: `https://youtu.be/gXlfXirQF3A`

2. Consider a parametrized path $\gamma(t)$ joining p_0 at $t = a$ to p_1 at $t = b$. Recall the length

$$L(\gamma) = \int_a^b |\dot{\gamma}_1|_g \, dt = \int_a^b \frac{ds}{dt} \, dt = \int_{p_0}^{p_1} \, ds.$$

That is the length of a curve is integral of infinitesimal line elements from the starting point p_0 pointing along the curve head-to-tail until they reach the endpoint p_1. That infinitesimality/limit admits a computation without reference to the parametrization.

Questions

1. none

Agenda: November 12, 2020

1. Algebra 2. Geometry 3. E&M 4. Teaching 5. Applied Boulder

Errata

1. p 18 "rough idea rough here"
2. p 23 "in [a] great circle"
3. p 24 "come form straight"
4. p 25 "with respect [to] any"
5. p 36 "point out ~~to~~ a consequence"
6. p 39 "2 000 years" (remove space)
7. p 41 "mentioned that [there] is another"